

Activité de redémarrage

1. Teste d'accès SSH à la VM à partir la machine physique :

```
arij@server26:~$ systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: active (running) since Tue 2026-09-29 21:12:39 UTC; 6min ago
   Invocation: 3d1599629ba34fe2b2cb15167555de4d
   TriggeredBy: ● ssh.socket
     Docs: man:sshd(8)
           man:sshd_config(5)
    Main PID: 1869 (sshd)
      Tasks: 1 (limit: 1426)
     Memory: 5.2M (peak: 7.1M)
        CPU: 145ms
     CGroup: /system.slice/ssh.service
            └─1869 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

sept. 29 21:12:39 server26 systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
sept. 29 21:12:39 server26 sshd[1869]: Server listening on 0.0.0.0 port 22.
sept. 29 21:12:39 server26 systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
sept. 29 21:12:39 server26 sshd[1869]: Server listening on :: port 22.
sept. 29 21:12:41 server26 sshd-session[1871]: Accepted password for arij from 192.168.159.1 port 58970 ssh2
sept. 29 21:12:41 server26 sshd-session[1871]: pam_unix(sshd:session): session opened for user arij(uid=1000) by arij(uid=0)
arij@server26:~$ ss -tlnp | grep :22
LISTEN 0      4096      0.0.0.0:22      0.0.0.0:*
LISTEN 0      4096      :::22          ::::*
arij@server26:~$
```

2. Sécurisation du SSH :

```
C:\Users\arijb>ssh-keygen -t ed25519 -C "arij@server26"
Generating public/private ed25519 key pair.
Enter file in which to save the key (C:\Users\arijb/.ssh/id_ed25519):
C:\Users\arijb/.ssh/id_ed25519 already exists.
Overwrite (y/n)? y
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in C:\Users\arijb/.ssh/id_ed25519
Your public key has been saved in C:\Users\arijb/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:A756EGgyM06KnqLCYyvCEcZVT8apxUd8xczcGxjG/ZY arij@server26
The key's randomart image is:
+--[ED25519 256]--+
| ..oo+. B+= |
| . . += o ..B + |
| . . . +. . . = |
|==o .o . . . Eo |
|**.. .. S . . |
|oo . . . . |
|+ o . . |
|B . . |
|O.o . . |
+-----[SHA256]-----+
C:\Users\arijb>
```

3. Puis Lancement la commande de copie et teste

```
C:\Users\arijb>type %USERPROFILE%.ssh\id_ed25519.pub | ssh arij@192.168.159.144 "mkdir -p ~/.ssh && chmod 700 ~/.ssh && cat >> ~/.ssh/authorized_keys && chmod 600 ~/.ssh/authorized_keys"

C:\Users\arijb>ssh arij@192.168.159.144
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of mar. 29 sept 2026 21:32:09 UTC

System load:  0.03          Processes:      226
Usage of /:   28.9% of 23.45GB Users logged in: 0
Memory usage: 12%          IPv4 address for ens33: 192.168.159.144
Swap usage:   0%

La maintenance de sécurité étendue pour Applications n'est pas activée.

134 mises à jour peuvent être appliquées immédiatement.
113 de ces mises à jour sont des mises à jour de sécurité.
Pour afficher ces mises à jour supplémentaires, exécuter : apt list --upgradable

Activez ESM Apps pour recevoir des futures mises à jour de sécurité supplémentaires.
Visitez https://ubuntu.com/esm ou exécutez : sudo pro status

Last login: Tue Sep 29 21:24:30 2026 from 192.168.159.1
arij@server26:~$
```

4. installer Docker

a) Dépendances :

```
arij@server26:~$ sudo apt install ca-certificates curl -y
ca-certificates est déjà la version la plus récente (20260601~26.04.1).
ca-certificates passé en « installé manuellement ».
curl est déjà la version la plus récente (8.18.0-1ubuntu2.7).
curl passé en « installé manuellement ».
Sommaire :
  Mise à niveau de : 0. Installation de : 0, Supprimé : 0. Non mis à jour : 0
arij@server26:~$
```

b) Clé GPG Docker :

```
arij@server26:~$ sudo install -m 0755 -d /etc/apt/keyrings
arij@server26:~$ sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
arij@server26:~$ sudo chmod a+r /etc/apt/keyrings/docker.asc
arij@server26:~$
```

c) Ajout du dépôt Docker :

```
arij@server26:~$ echo \
"deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.asc] https://download.docker.com/linux/ubuntu \
$(. /etc/os-release && echo "$VERSION_CODENAME") stable" | \
sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
arij@server26:~$ cat /etc/apt/sources.list.d/docker.list
deb [arch=amd64 signed-by=/etc/apt/keyrings/docker.asc] https://download.docker.com/linux/ubuntu   resolute stable
```

d) Mise à jour + installation de Docker :

```
arij@server26:~$ sudo apt update
Atteint : 1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Atteint : 2 http://tn.archive.ubuntu.com/ubuntu resolute InRelease
Atteint : 3 http://tn.archive.ubuntu.com/ubuntu resolute-updates InRelease
Réception de : 4 https://download.docker.com/linux/ubuntu resolute InRelease [32,5 kB]
Atteint : 5 http://tn.archive.ubuntu.com/ubuntu resolute-backports InRelease
Réception de : 6 https://download.docker.com/linux/ubuntu resolute/stable amd64 Packages [26,8 kB]
59,3 ko réceptionnés en 1s (118 ko/s)
Tous les paquets sont à jour.
```

```
arij@server26:~$ sudo apt install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin -y
Installation de :
  containerd.io  docker-buildx-plugin  docker-ce  docker-ce-cli  docker-compose-plugin

Installation de dépendances :
  docker-ce-rootless-extras  pigz

Paquets suggérés :
  cgroupfs-mount | cgroup-lite  docker-model-plugin

Sommaire :
  Mise à niveau de : 0. Installation de : 7, Supprimé : 0. Non mis à jour : 0
Taille du téléchargement : 101 Mo
Espace nécessaire : 383 Mo / 15,9 Go disponible

Réception de : 1 https://download.docker.com/linux/ubuntu resolute/stable amd64 containerd.io amd64 2.3.6-1~ubuntu.26.04~resolute [23,2 MB]
Réception de : 2 http://tn.archive.ubuntu.com/ubuntu resolute/universe amd64 pigz amd64 2.8-1build1 [66,8 kB]
Réception de : 3 https://download.docker.com/linux/ubuntu resolute/stable amd64 docker-ce-cli amd64 5:29.8.1-1~ubuntu.26.04~resolute [17,6 MB]
Réception de : 4 https://download.docker.com/linux/ubuntu resolute/stable amd64 docker-ce amd64 5:29.8.1-1~ubuntu.26.04~resolute [24,3 MB]
Réception de : 5 https://download.docker.com/linux/ubuntu resolute/stable amd64 docker-buildx-plugin amd64 0.37.1-1~ubuntu.26.04~resolute [17,3 MB]
Réception de : 6 https://download.docker.com/linux/ubuntu resolute/stable amd64 docker-ce-rootless-extras amd64 5:29.8.1-1~ubuntu.26.04~resolute [10,2 MB]
Réception de : 7 https://download.docker.com/linux/ubuntu resolute/stable amd64 docker-compose-plugin amd64 5.5.1-1~ubuntu.26.04~resolute [8 013 kB]
101 Mo réceptionnés en 20s (5 112 ko/s)
```

e) Vérifier que le service tourne :

```
arij@server26:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-09-29 21:58:59 UTC; 1min 46s ago
   Invocation: fe469debe3654fc4a009f8f832223dd5
   TriggeredBy: ● docker.socket
     Docs: https://docs.docker.com
    Main PID: 2543 (dockerd)
      Tasks: 9
     Memory: 31.3M (peak: 31.6M)
        CPU: 1.057s
    CGroup: /system.slice/docker.service
            └─2543 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

sept. 29 21:58:58 server26 dockerd[2543]: time="2026-09-29T21:58:58.122536925Z" level=info msg="Restoring containers: s>
sept. 29 21:58:58 server26 dockerd[2543]: time="2026-09-29T21:58:58.184067911Z" level=info msg="Deleting nftables IPv4 >
```

f) Autorisation à utiliser Docker sans sudo :

```
arij@server26:~$ sudo usermod -aG docker $USER
arij@server26:~$
```

g) Se déconnecter puis se reconnecter :

```
arij@server26:~$ exit
logout
Connection to 192.168.159.144 closed.

C:\Users\arijb>ssh arij@192.168.159.144
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-34-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of mar. 29 sept 2026 22:03:14 UTC

System load:  0.0               Processes:            225
Usage of /:   33.3% of 23.45GB   Users logged in:     0
Memory usage: 14%              IPv4 address for ens33: 192.168.159.144
Swap usage:   0%

La maintenance de sécurité étendue pour Applications n'est pas activée.

0 mise à jour peut être appliquée immédiatement.

Activez ESM Apps pour recevoir des futures mises à jour de sécurité supplémentaires.
Visitez https://ubuntu.com/esm ou exécutez : sudo pro status

Last login: Tue Sep 29 21:52:18 2026 from 192.168.159.1
```

h) Test final :

```
arij@server26:~$ docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:5e23090353324d887c48ad5e5c56d294eab81588df9605b07d1afe895f9cc8f8
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/
```

5.la finalisation de la sécurisation :

```
arij@server26:~$ sudo nano /etc/ssh/sshd_config.d/00-hardening.conf
[sudo: authenticate] Password:
arij@server26:~$ sudo systemctl reload ssh
sudo sshd -T | grep -E "passwordauthentication|permitrootlogin|pubkeyauthentication"
passwordauthentication no
pubkeyauthentication yes
passwordauthentication no
arij@server26:~$
```

```
C:\Users\arijb>ssh arij@192.168.159.144
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-34-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of mar. 29 sept 2026 22:13:07 UTC

System load:  0.0           Processes:           225
Usage of /:   33.3% of 23.45GB Users logged in:         0
Memory usage: 14%          IPv4 address for ens33: 192.168.159.144
Swap usage:   0%

La maintenance de sécurité étendue pour Applications n'est pas activée.

0 mise à jour peut être appliquée immédiatement.

Activez ESM Apps pour recevoir des futures mises à jour de sécurité supplémentaires.
Visitez https://ubuntu.com/esm ou exécutez : sudo pro status

Last login: Tue Sep 29 22:11:20 2026 from 192.168.159.1
arij@server26:~$ ssh -o PreferredAuthentications=password -o PubkeyAuthentication=no arij@192.168.159.144
arij@192.168.159.144: Permission denied (publickey).
arij@server26:~$
```

5- Installez Jenkins en tant que service et vérifiez son bon fonctionnement à partir de la machine physique :

a. Installer Java :

```
arij@server26:~$ sudo apt install -y fontconfig openjdk-21-jre
Installation de :
  fontconfig  openjdk-21-jre

Installation de dépendances :
  adwaita-icon-theme      libatk-wrapper-java      libglx-mesa0             librsvg2-2               libxfixes3
  alsa-topology-conf      libatk-wrapper-java-jni  libglx0                  librsvg2-common          libxft2
  alsa-ucm-conf           libatk1.0-0t64           libglycin-2-0            libsharpyuv0             libxi6
  at-spi2-common          libatspi2.0-0t64         libgraphite2-3           libsm6                   libxinerama1
  at-spi2-core            libavahi-client3         libgtk-3-0t64            libthai-data             libxkbfile1
  bubblewrap              libavahi-common-data     libgtk-3-bin             libthai0                 libxmu6
  ca-certificates-java    libavahi-common3         libgtk-3-common          libvulkan1               libxpm4
  dconf-gsettings-backend libcairo-gobject2        libharfbuzz0b            libwayland-client0       libxrandr2
  dconf-service           libcairo2                libheif-plugin-aomdec    libwayland-cursor0      libxrender1
  fontconfig-config       libcolord2               libheif-plugin-aomenc    libwayland-egl1         libxres1
  fonts-dejavu-core       libcups2t64              libheif1                 libx11-xcb1             libxshmfence1
  fonts-dejavu-extra      libdatriel               libhwylt64               libxaw7                  libxt6t64
  fonts-dejavu-mono       libdconf1                libibus-1.0-5            libxcb-dri3-0            libxtst6
  glycin-loaders          libdisplay-info3         libice6                  libxcb-glx0              libxv1
  glycin-thumbnails       libdrm-intel1            libjpeg-turbo8           libxcb-present0          libxxf86dgal
  gsettings-desktop-schemas libepoxy0                libjpeg8                 libxcb-randr0            libxxf86vm1
  gtk-update-icon-cache   libfontconfig1           libjxl0.11               libxcb-render0           luit
  hicolor-icon-theme      libgbm1                  liblcms2-2               libxcb-shape0            mesa-libgallium
  ibus-gtk3              libgdk-pixbuf-2.0-0      libpango-1.0-0           libxcb-shm0              mesa-vulkan-drivers
  java-common             libgdk-pixbuf2.0-common  libpangocairo-1.0-0     libxcb-sync1             openjdk-21-jre-headless
  libao3                  libgif7                  libpangoft2-1.0-0        libxcb-xfixes0           user-session-migration
  libasound2-data         libglib1                 libpciaccess0            libxcomposite1           x11-common
  libasound2t64           libglib-mesa-dri         libpcsclite1             libxcursor1              x11-utils
  libatk-bridge2.0-0t64   libglvnd0                libpixman-1-0            libxdamage1

arij@server26:~$ java -version
openjdk version "21.0.12.1" 2026-08-18
OpenJDK Runtime Environment (build 21.0.12.1+1-1-26.04.4-Ubuntu)
OpenJDK 64-Bit Server VM (build 21.0.12.1+1-1-26.04.4-Ubuntu, mixed mode, sharing)
arij@server26:~$
```

B .Ajouter le dépôt Jenkins + installer

Clé GPG Jenkins

```
arij@server26:~$ sudo wget -O /etc/apt/keyrings/jenkins-keyring.asc \
https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
--2026-09-30 09:23:00-- https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
Resolving pkg.jenkins.io (pkg.jenkins.io)... 199.232.142.133, 2a04:4e42:63::645
Connecting to pkg.jenkins.io (pkg.jenkins.io)[199.232.142.133]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1680 (1.6K) [application/octet-stream]
Saving to: '/etc/apt/keyrings/jenkins-keyring.asc'

/etc/apt/keyrings/jenkins-keyring 100%[=====>] 1,64K --.-KB/s in 0s

2026-09-30 09:23:01 (3,87 MB/s) - '/etc/apt/keyrings/jenkins-keyring.asc' saved [1680/1680]
```

c. Dépôt Jenkins

```
arij@server26:~$ echo "deb [signed-by=/etc/apt/keyrings/jenkins-keyring.asc] \
https://pkg.jenkins.io/debian-stable binary/ | sudo tee \
/etc/apt/sources.list.d/jenkins.list > /dev/null"
```

d. Installation

```
/etc/apt/sources.list.d/jenkins.list > /dev/null
arij@server26:~$ sudo apt update
Atteint : 1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Atteint : 2 http://tn.archive.ubuntu.com/ubuntu resolute InRelease
Atteint : 3 http://tn.archive.ubuntu.com/ubuntu resolute-updates InRelease
Ign : 4 https://pkg.jenkins.io/debian-stable binary/ InRelease
Atteint : 5 http://tn.archive.ubuntu.com/ubuntu resolute-backports InRelease
Atteint : 6 https://download.docker.com/linux/ubuntu resolute InRelease
Réception de : 7 https://pkg.jenkins.io/debian-stable binary/ Release [2 044 B]
Réception de : 8 https://pkg.jenkins.io/debian-stable binary/ Release.gpg [833 B]
Réception de : 9 https://pkg.jenkins.io/debian-stable binary/ Packages [31,8 kB]
34,7 ko réceptionnés en 2s (21,6 ko/s)
Tous les paquets sont à jour.
```

```
arij@server26:~$ sudo apt install -y jenkins
Installation de :
jenkins

Installation de dépendances :
net-tools

Sommaire :
Mise à niveau de : 0. Installation de : 2, Supprimé : 0. Non mis à jour : 0
Taille du téléchargement : 53,9 Mo
Espace nécessaire : 54,7 Mo / 14,9 Go disponible

Réception de : 1 http://tn.archive.ubuntu.com/ubuntu resolute/main amd64 net-tools amd64 2.10-2ubuntu1 [151 kB]
Réception de : 2 https://pkg.jenkins.io/debian-stable binary/ jenkins 2.580.1 [53,7 MB]
53,9 Mo réceptionnés en 1min 27s (620 ko/s)
Sélection du paquet net-tools précédemment désélectionné.
(Lecture de la base de données... 144139 fichiers et répertoires déjà installés.)
Préparation du dépaquetage de .../net-tools_2.10-2ubuntu1_amd64.deb ...
Dépaquetage de net-tools (2.10-2ubuntu1) ...
Sélection du paquet jenkins précédemment désélectionné.
Préparation du dépaquetage de .../jenkins_2.580.1_all.deb ...
Dépaquetage de jenkins (2.580.1) ...
Paramétrage de net-tools (2.10-2ubuntu1) ...
Paramétrage de jenkins (2.580.1) ...
Created symlink '/etc/systemd/system/multi-user.target.wants/jenkins.service' → '/usr/lib/systemd/system/jenkins.service'.
Traitement des actions différées (« triggers ») pour man-db (2.13.1-1build1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.
```

e. Démarrer + activer le service

```
arij@server26:~$ sudo systemctl start jenkins
arij@server26:~$ sudo systemctl enable jenkins
Synchronizing state of jenkins.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable jenkins
arij@server26:~$ sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-09-30 09:27:23 UTC; 1min 40s ago
  Invocation: bcee21822e1b4b98a7fafd840ef22360
    Main PID: 4987 (java)
      Tasks: 43 (limit: 3937)
     Memory: 552.2M (peak: 567.5M)
        CPU: 36.414s
    CGroup: /system.slice/jenkins.service
            └─4987 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpP

sept. 30 09:26:23 server26 jenkins[4987]: [LF]> This may also be found at: /var/lib/jenkins/secrets/initialAdminPassword
sept. 30 09:26:23 server26 jenkins[4987]: [LF]>
sept. 30 09:26:23 server26 jenkins[4987]: [LF]> *****
sept. 30 09:26:23 server26 jenkins[4987]: [LF]> *****
sept. 30 09:26:23 server26 jenkins[4987]: [LF]> *****
sept. 30 09:27:23 server26 jenkins[4987]: 2026-09-30 09:27:23.446+0000 [id=40] INFO jenkins.InitReactorRunner$1#onAtta>
sept. 30 09:27:23 server26 jenkins[4987]: 2026-09-30 09:27:23.480+0000 [id=30] INFO hudson.lifecycle.Lifecycle#onReady>
sept. 30 09:27:23 server26 systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
sept. 30 09:27:25 server26 jenkins[4987]: 2026-09-30 09:27:25.482+0000 [id=56] INFO h.m.DownloadService$Downloadable#l>
sept. 30 09:27:25 server26 jenkins[4987]: 2026-09-30 09:27:25.488+0000 [id=56] INFO hudson.util.Retrier#start: Perform>
arij@server26:~$
```

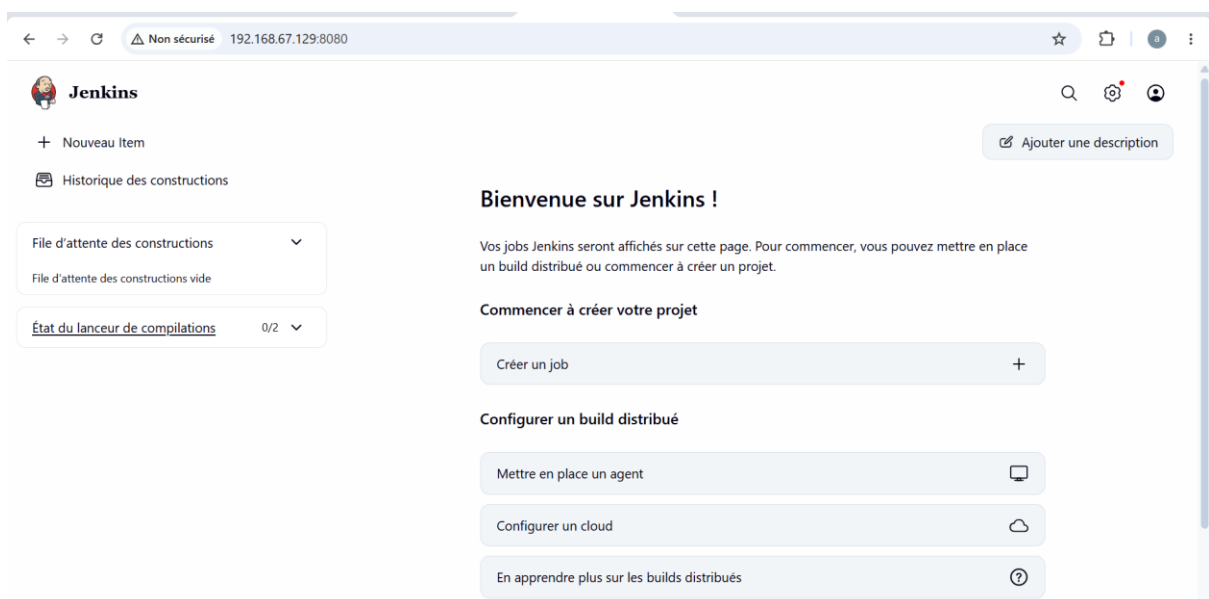
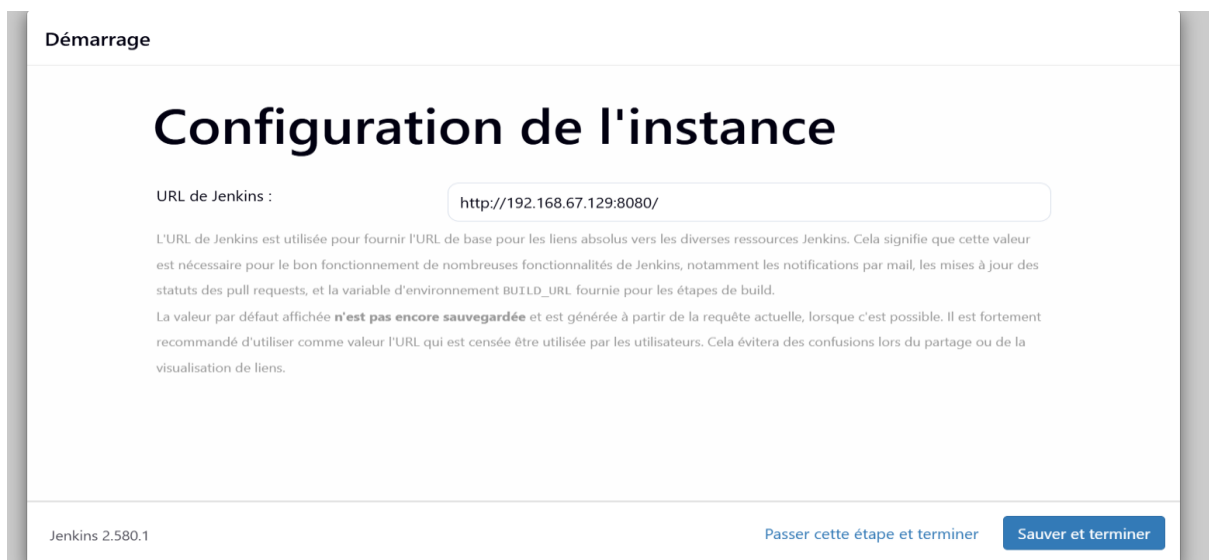
f. Ouvrir le port 8080 dans le pare-feu

```
arij@server26:~$ sudo ufw allow 8080/tcp
Rules updated
Rules updated (v6)
```

g. Récupérer le mot de passe initial

```
arij@server26:~$ sudo cat /var/lib/jenkins/secrets/initialAdminPassword
993e59e624be4b1bacea6ff67ee9917
arij@server26:~$
```

h. Test depuis Windows



6. Créez un mini CV One Page (HTML5/CSS3/Javascript) dans un dossier géré avec git et fournir son lien GitHub.

a. Créer un dossier pour le projet :

```
arij@server26:~$ mkdir -p ~/cv-onepage
arij@server26:~$ cd ~/cv-onepage
arij@server26:~/cv-onepage$
```

B.vérifier git est installé :

```
arij@server26:~/cv-onepage$ git --version
git version 2.53.0
```

C.Configurer Git :

```
arij@server26:~/cv-onepage$ git config --global user.name "arijbelaid"
arij@server26:~/cv-onepage$ git config --global user.email "aarijbelaid@gmail.com"
arij@server26:~/cv-onepage$ git config --list
user.name=arijbelaid
user.email=aarijbelaid@gmail.com
arij@server26:~/cv-onepage$
```

d.Créer les fichiers du CV :

```
arij@server26:~/cv-onepage$ nano index.html
arij@server26:~/cv-onepage$ nano style.css
arij@server26:~/cv-onepage$ nano script.js
```

Index.html

```
GNU nano 8.7.1 index.html *
<!DOCTYPE html>
<html lang="fr">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>CV - Arij</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <header>
    <h1>Arij [Nom]</h1>
    <p class="subtitle">Étudiant(e) en Informatique</p>
  </header>
  <main>
    <section id="about">
      <h2>À propos</h2>
      <p>Passionné(e) par le DevOps, les systèmes Linux et l'automatisation.</p>
    </section>
    <section id="skills">
      <h2>Compétences</h2>
      <ul>
        <li>Linux / Ubuntu Server</li>
        <li>Docker</li>
        <li>Jenkins (CI/CD)</li>
        <li>Git / GitHub</li>
        <li>HTML5 / CSS3 / JavaScript</li>
      </ul>
    </section>
  </main>
</body>
</html>
```

Style.css

```
GNU nano 8.7.1 style.css
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
  font-family: 'Segoe UI', Arial, sans-serif;
  line-height: 1.6;
  color: #333;
  background-color: #f4f4f4;
  padding: 20px;
}

header {
  background: linear-gradient(135deg, #667eea, #764ba2);
  color: white;
  padding: 40px 20px;
  border-radius: 10px;
  text-align: center;
  margin-bottom: 20px;
}

header h1 {
  font-size: 2.5em;
  margin-bottom: 10px;
}

.subtitle {
  font-size: 1.2em;
  margin-top: 5px;
}
```

Script.js

```
GNU nano 8.7.1 script.js
// Animation d'apparition des sections au scroll
document.addEventListener('DOMContentLoaded', () => {
  const sections = document.querySelectorAll('section');

  const observer = new IntersectionObserver((entries) => {
    entries.forEach(entry => {
      if (entry.isIntersecting) {
        entry.target.style.opacity = '1';
        entry.target.style.transform = 'translateY(0)';
      }
    });
  }, { threshold: 0.1 });

  sections.forEach(section => {
    section.style.opacity = '0';
    section.style.transform = 'translateY(20px)';
    section.style.transition = 'opacity 0.6s ease, transform 0.6s ease';
    observer.observe(section);
  });

  // Affichage de l'année dynamique
  const year = new Date().getFullYear();
  document.querySelector('footer p').innerHTML = `&copy; ${year} Arij. Tous droits réservés.`;
});
```

Voici cv

The screenshot shows a resume website for Arij Belaid, a DevOps & Cloud Engineer in training. The website is dark-themed and displays various sections including profile, education, competencies, projects, and certifications.

Profile: Arij Belaid, Étudiante en Master Pro DevOps & Cloud, Titulaire d'une Licence IT — Développement de Systèmes d'Information. Contact: votre.email@example.com, github.com/VOTRE_USERNAME, LinkedIn, Tunisie.

Formation:

- Master Professionnel — DevOps & Cloud (2026 - en cours): Spécialisation : CI/CD, conteneurisation, orchestration, cloud (AWS/Azure), IaC.
- Licence IT — Développement de Systèmes d'Information (2023 - 2026): Bases solides en programmation, bases de données, génie logiciel et réseaux.

Compétences:

- DEVOPS & CLOUD: Docker, Jenkins, Git / GitHub, CI/CD, Linux (Ubuntu Server), Bash.
- DEVELOPPEMENT: HTML5 / CSS3, JavaScript, Python, Java, SQL / MySQL, PHP.
- OUTILS: VMware, VS Code, Git Bash, UFW, OpenSSH.

Projets & Expérience:

- Mise en place d'une chaîne CI/CD complète (Projet académique - 2026): Installation et durcissement d'un serveur Ubuntu Server 26.04, Configuration d'un accès distant sécurisé par clé SSH ED25519, Déploiement de Docker et de Jenkins en services systemd, Ouverture maîtrise du pare-feu UFW (SSH + Jenkins).
- Mini CV One Page — HTML5 / CSS3 / JS (Projet personnel - 2026): Conception d'un site responsive en pur HTML/CSS/JS, Versionné avec Git et publié sur GitHub Pages.
- Développement d'une application de gestion (Projet Licence - 2025): Application web avec base de données MySQL, Front-end en HTML/CSS/JS, back-end en PHP.

Certifications & Veille:

- Auto-formation continue sur Docker et Jenkins.
- Lecture active de la documentation Linux Foundation.
- Suivi des bonnes pratiques DevOps (The Twelve-Factor App).

© 2026 Arij Belaid - CV One Page réalisé en HTML5 / CSS3 / JavaScript

6- Activer les push GitHub via SSH : Ouvrez GitHub / Settings → SSH and GPG keys / Cliquez sur New SSH key / Donnez un nom à la clé / Collez votre clé publique / Enregistrez / Tester: ssh -T git@github.com, Configurer le dépôt local pour utiliser SSH.

a.Générer et ajouter la clé SSH sur votre VM : Vérifier les clés existantes

```
arij@server26:~/cv-onepage$ ls -al ~/.ssh
total 20
drwx----- 2 arij arij 4096 Sep 29 21:29 .
drwxr-x--- 6 arij arij 4096 Sep 30 10:00 ..
-rw----- 1 arij arij 192 Sep 29 21:31 authorized_keys
-rw----- 1 arij arij 978 Sep 29 21:29 known_hosts
-rw-r--r-- 1 arij arij 142 Sep 29 21:29 known_hosts.old
arij@server26:~/cv-onepage$
```

b.Générer une nouvelle paire de clés SSH :

```
arij@server26:~/cv-onepage$ ssh-keygen -t ed25519 -C "aarijbelaidd@gmail.com"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/arij/.ssh/id_ed25519):
Enter passphrase for "/home/arij/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/arij/.ssh/id_ed25519
Your public key has been saved in /home/arij/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:nmhBXDM62NnfrRGxr0q+hx2LnSvJ/siUE7hsJ9RGkQo aarijbelaidd@gmail.com
The key's randomart image is:
+--[ED25519 256]--+
|      + ...      |
|    +E= o.. o    |
|  . B... o       |
|   ..= . +       |
|  . S = o o      |
|   * + o.+       |
|  o B.**o+        |
|   . B**+        |
|    .B=o.        |
+-----[SHA256]-----+
arij@server26:~/cv-onepage$
```

c.Afficher la clé publique :

```
arij@server26:~/cv-onepage$ cat ~/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIIfeNBGqKovV1Z1ttYTiE18Jf2jPGyzG8c+cJ2qhPJmL aarijbelaidd@gmail.com
arij@server26:~/cv-onepage$
```

d.Tester la connexion SSH :

```
arij@server26:~/cv-onepage$ ssh -T git@github.com
The authenticity of host 'github.com (140.82.121.3)' can't be established.
ED25519 key fingerprint is: SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvC0qu
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
Hi arijbelaidd! You've successfully authenticated, but GitHub does not provide shell access.
arij@server26:~/cv-onepage$
```

e.Configurer le dépôt local pour utiliser SSH

```
arij@server26:~/cv-onepage$ git remote set-url origin git@github.com:arijbelaidd/cv-onepage.git
arij@server26:~/cv-onepage$ git remote -v
origin  git@github.com:arijbelaidd/cv-onepage.git (fetch)
origin  git@github.com:arijbelaidd/cv-onepage.git (push)
arij@server26:~/cv-onepage$ nano ~/cv-onepage/README.md
arij@server26:~/cv-onepage$ git add .
arij@server26:~/cv-onepage$ git commit -m "docs: ajout README avec commandes SSH et capture"
[main a05a1e6] docs: ajout README avec commandes SSH et capture
1 file changed, 76 insertions(+)
create mode 100644 README.md
arij@server26:~/cv-onepage$ git push -u origin main
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 2 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (8/8), 6.31 KiB | 538.00 KiB/s, done.
Total 8 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To github.com:arijbelaidd/cv-onepage.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
arij@server26:~/cv-onepage$
```

Évolution du mini-CV vers DevSecOps Portfolio

7- Faites évoluer le mini CV en une petite application DevSecOps Portfolio comportant au minimum les sections : About / Skills / Projects / Experience / Contact.

README.md : ajoutez une capture de la nouvelle version et décrivez brièvement les principales améliorations.

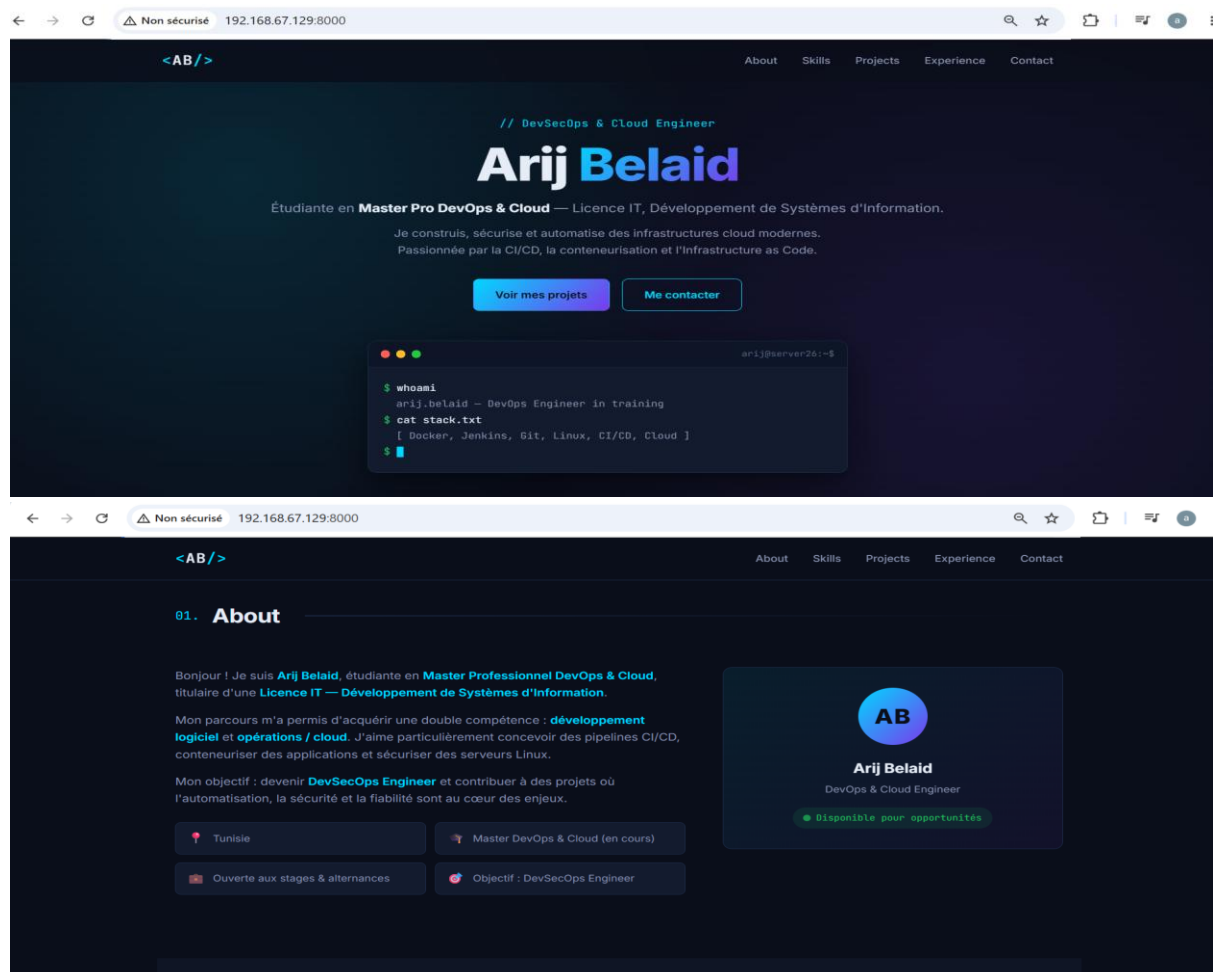
a. Réécrire index.html ,style.css et script.js en version DevSecOps Portfolio

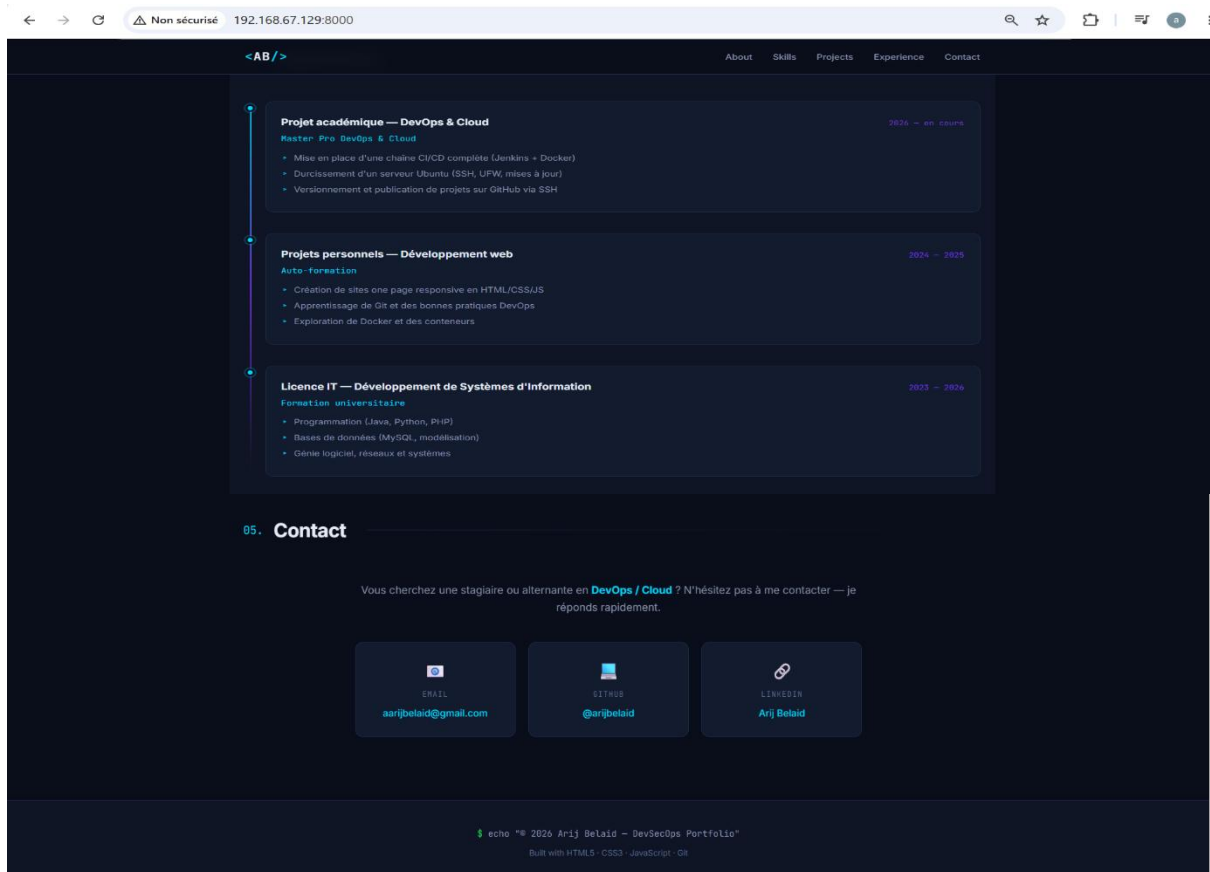
```
arij@server26:~/cv-onepage$ nano index.html
arij@server26:~/cv-onepage$ nano style.css
arij@server26:~/cv-onepage$ nano script.js
arij@server26:~/cv-onepage$ nano README.md
```

b. Lancer en local

```
arij@server26:~/cv-onepage$ git clone git@github.com:arijbelaidd/cv-onepage.git
Cloning into 'cv-onepage'...
remote: Enumerating objects: 8, done.
remote: Counting objects: 100% (8/8), done.
remote: Compressing objects: 100% (7/7), done.
remote: Total 8 (delta 1), reused 8 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (8/8), 6.31 KiB | 3.16 MiB/s, done.
Resolving deltas: 100% (1/1), done.
arij@server26:~/cv-onepage$ python3 -m http.server 8000
Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...
192.168.67.1 - - [03/Oct/2026 21:12:55] "GET / HTTP/1.1" 200 -
192.168.67.1 - - [03/Oct/2026 21:12:55] "GET /style.css HTTP/1.1" 200 -
192.168.67.1 - - [03/Oct/2026 21:12:55] "GET /script.js HTTP/1.1" 200 -
```

Cv :





c. Transférer la capture dans la VM :

```
C:\Users\arijb>scp %USERPROFILE%\Desktop\screenshot.png arij@192.168.67.129:~/cv-onepage/screenshot.png
screenshot.png                                100% 474KB 46.3MB/s   00:00
```

```
arij@server26:~/cv-onepage$ ls -la ~/cv-onepage/screenshot.png
-rw-rw-r-- 1 arij arij 485435 Oct  3 21:32 /home/arij/cv-onepage/screenshot.png
arij@server26:~/cv-onepage$ file ~/cv-onepage/screenshot.png
/home/arij/cv-onepage/screenshot.png: PNG image data, 1884 x 913, 8-bit/color RGBA, non-interlaced
arij@server26:~/cv-onepage$
```

d. Commit + Push

```
arij@server26:~/cv-onepage$ git push
Enumerating objects: 12, done.
Counting objects: 100% (12/12), done.
Delta compression using up to 2 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (7/7), 469.99 KiB | 3.38 MiB/s, done.
Total 7 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:arijbelaidd/cv-onepage.git
 a05ale6..f57ce9e main -> main
arij@server26:~/cv-onepage$
```

e/ajout de Démo en ligne

<https://arijbelaidd.github.io/cv-onepage/>

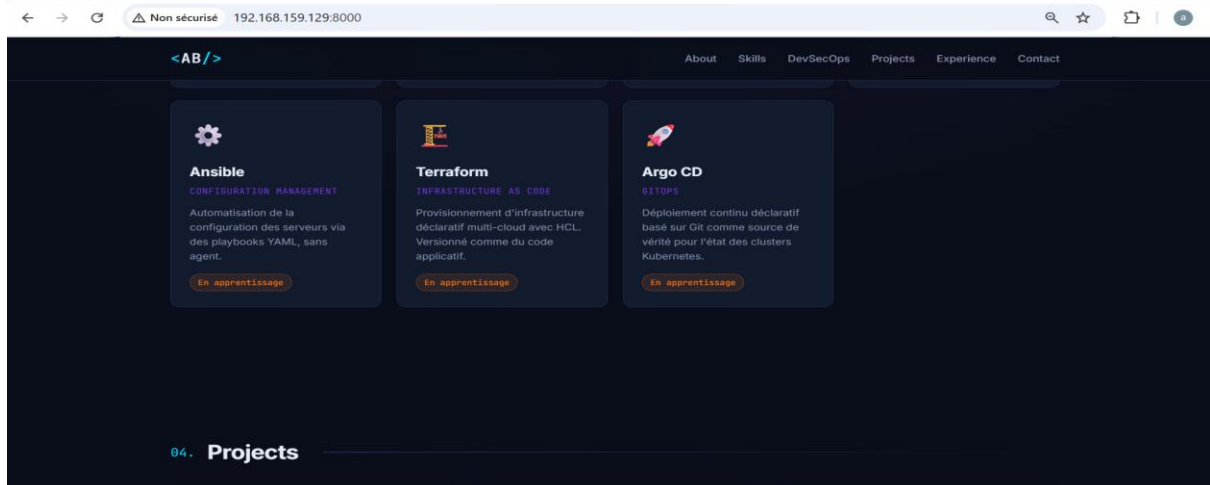
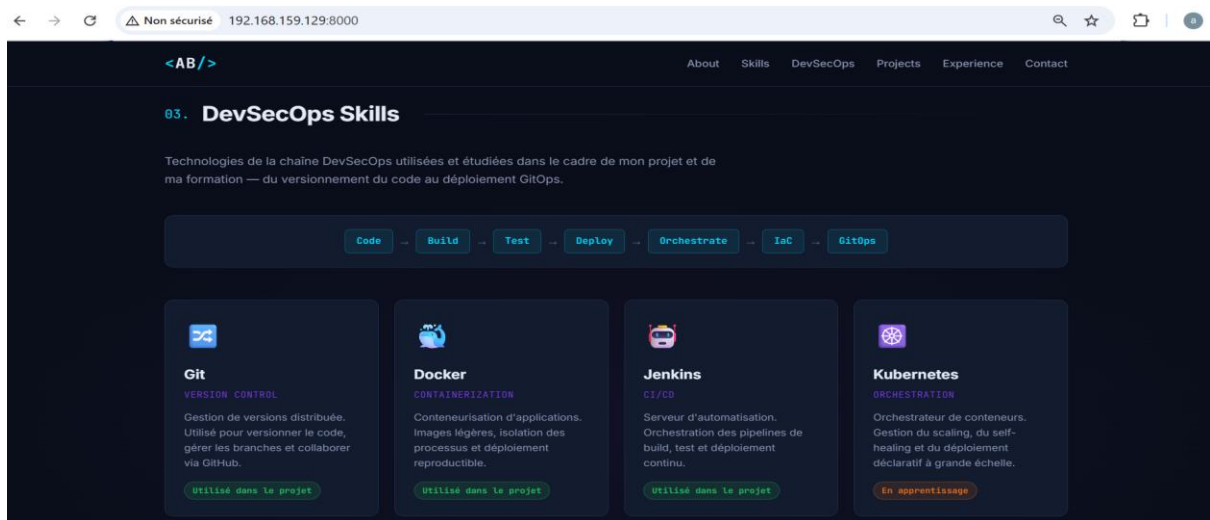
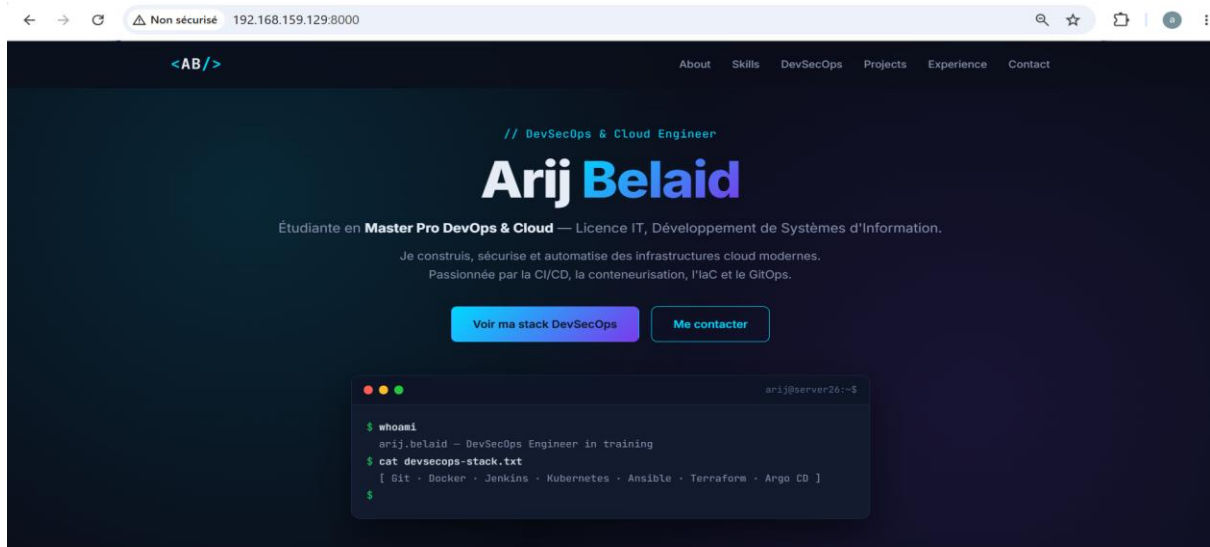
8- Ajoutez une section DevSecOps Skills permettant d'afficher les technologies utilisées pendant le projet : Git / Docker / Jenkins / Kubernetes / Ansible / Terraform / Argo CD

README.md : ajoutez une capture d'écran.

a. Réécrire index.html ,style.css et script.js :

```
arij@server26:~/cv-onepage$ nano index.html
arij@server26:~/cv-onepage$ nano style.css
arij@server26:~/cv-onepage$ nano script.js
```

b.cv partie ajoutée DevSecOps :



*Transférer la capture de partie DevSecOps vers la VM :

```
C:\Users\arijb>scp %USERPROFILE%\Desktop\screenshot1.png arij@192.168.159.129:~/cv-onepage/screenshot1.png
screenshot1.png                                100% 229KB 18.7MB/s   00:00
C:\Users\arijb>
```

```
arij@server26:~/cv-onepage$ ls -la ~/cv-onepage/screenshot1.png
-rw-rw-r-- 1 arij arij 234824 Oct  6 08:51 /home/arij/cv-onepage/screenshot1.png
```

c. Commit + Push :

```
arij@server26:~/cv-onepage$ git add .
arij@server26:~/cv-onepage$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   README.md
        modified:   index.html
        new file:   screenshot1.png
        modified:   script.js
        modified:   style.css

arij@server26:~/cv-onepage$ git commit -m "feat(v3): ajout section DevSecOps Skills (Git/Docker/Jenkins/K8s/Ansible/Terraform/Argo CD) + capture"
[main 1a281bf] feat(v3): ajout section DevSecOps Skills (Git/Docker/Jenkins/K8s/Ansible/Terraform/Argo CD) + capture
5 files changed, 364 insertions(+), 76 deletions(-)
create mode 100644 screenshot1.png
arij@server26:~/cv-onepage$ git push
Enumerating objects: 12, done.
Counting objects: 100% (12/12), done.
Delta compression using up to 2 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (7/7), 222.21 KiB | 1.90 MiB/s, done.
Total 7 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
To github.com:arijbelaidd/cv-onepage.git
   f57ce9e..1a281bf  main -> main
arij@server26:~/cv-onepage$
```

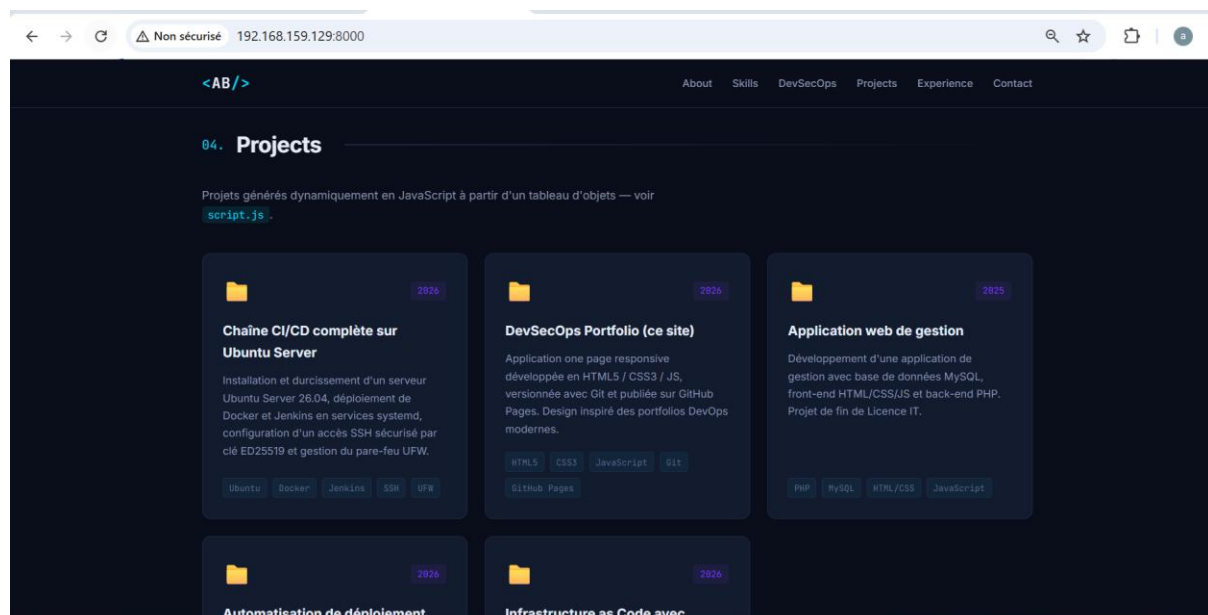
9- Ajoutez une section Projects générée dynamiquement en JavaScript à partir d'un tableau d'objets.

README.md : ajoutez un extrait du code JavaScript et une capture du résultat.

a. Réécrire index.html ,style.css et script.js :

```
arij@server26:~/cv-onepage$ nano index.html
arij@server26:~/cv-onepage$ nano script.js
arij@server26:~/cv-onepage$ nano style.css
```

b. Tester en local :



c. Commit + Push :

```
arij@server26:~/cv-onepage$ cat ~/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIOS5flwXHyYq090n4HRPoHUCfX1No//Xyj74DHpxyiIJ aarijbelaidd@gmail.com
arij@server26:~/cv-onepage$ ssh -T git@github.com
Hi arijbelaidd! You've successfully authenticated, but GitHub does not provide shell access.
arij@server26:~/cv-onepage$ git push
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 2 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 4.74 KiB | 606.00 KiB/s, done.
Total 6 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To github.com:arijbelaidd/cv-onepage.git
   1a281bf..10d1e68  main -> main
arij@server26:~/cv-onepage$
```

Dockerisation initiale :

10- Créez un Dockerfile pour servir votre DevSecOps Portfolio avec Nginx.
README.md : ajoutez le contenu du Dockerfile et une courte explication.

a. Créer Dockerfile,nginx.conf et Dockerignore

```
arij@server26:~/cv-onepage$ nano Dockerfile
arij@server26:~/cv-onepage$ nano nginx.conf
arij@server26:~/cv-onepage$ nano .dockerignore
```

b.Construire l'image Docker :

```
arij@server26:~/cv-onepage$ docker build -t cv-onepage:1.0 .
[+] Building 19.8s (12/12) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.1s
=> => transferring dockerfile: 1.09kB                             0.0s
=> [internal] load metadata for docker.io/library/nginx:alpine    3.4s
=> [internal] load .dockerignore                                  0.1s
=> => transferring context: 128B                                     0.0s
=> [1/7] FROM docker.io/library/nginx:alpine@sha256:df221db836e1754089190208cee7eeda94f233197056426eda74a43ab1abeac2 10.8s
=> => resolve docker.io/library/nginx:alpine@sha256:df221db836e1754089190208cee7eeda94f233197056426eda74a43ab1abeac2 0.2s
=> => sha256:e72112c14215406dc7f23ef8e5cfd8deb8662176167382ad64ec86f52b7a88e 20.53MB / 20.53MB 4.6s
=> => sha256:64c8194480fee2c03e01cc561cf5542733627d2c7de493a028139677025f975c 1.21kB / 1.21kB 1.6s
=> => sha256:e76228b47809e8a4bd37a6ba52a04d282d8c72ddf0ff9e8b83fd6318631abd52 1.40kB / 1.40kB 1.0s
=> => sha256:9a9a644fdd6a3681d43dace44c8cfef374cfd7b24f433113379ea483692498a8 404B / 404B 1.0s
=> => sha256:745dfb2690dd8627e1593941cacc756f14943df7930a478beb4e808c5695cb9 965B / 965B 0.4s
=> => sha256:6c53d0b2a666ed8f8fad8d9982e30cf24eb085036c17b27591665ca25e1ba40c 629B / 629B 0.4s
=> => sha256:d9aae54b58315e84b9c3ac5f00bad6610e0e28ad9ff28fcde694647e0be13131 1.93MB / 1.93MB 2.8s
=> => sha256:e2de96513ba9eb53b431787ec8a65cddc380ac4772a3e4c4b714dcfde2a102b5 3.85MB / 3.85MB 4.1s
=> => extracting sha256:e2de96513ba9eb53b431787ec8a65cddc380ac4772a3e4c4b714dcfde2a102b5 0.9s
=> => extracting sha256:d9aae54b58315e84b9c3ac5f00bad6610e0e28ad9ff28fcde694647e0be13131 0.7s
=> => extracting sha256:6c53d0b2a666ed8f8fad8d9982e30cf24eb085036c17b27591665ca25e1ba40c 0.0s
=> => extracting sha256:745dfb2690dd8627e1593941cacc756f14943df7930a478beb4e808c5695cb9 0.0s
=> => extracting sha256:9a9a644fdd6a3681d43dace44c8cfef374cfd7b24f433113379ea483692498a8 0.0s
=> => extracting sha256:64c8194480fee2c03e01cc561cf5542733627d2c7de493a028139677025f975c 0.0s
=> => extracting sha256:e76228b47809e8a4bd37a6ba52a04d282d8c72ddf0ff9e8b83fd6318631abd52 0.0s
=> => extracting sha256:e72112c14215406dc7f23ef8e5cfd8deb8662176167382ad64ec86f52b7a88e 3.0s
=> [internal] load build context                                0.2s
=> => transferring context: 534.99kB                             0.1s
=> [2/7] RUN rm -rf /usr/share/nginx/html/*                    3.2s
=> [3/7] COPY index.html /usr/share/nginx/html/                0.3s
=> [4/7] COPY style.css /usr/share/nginx/html/                  0.1s
=> [5/7] COPY script.js /usr/share/nginx/html/                  0.0s
=> [6/7] COPY screenshot.png /usr/share/nginx/html/             0.0s
=> [7/7] COPY nginx.conf /etc/nginx/conf.d/default.conf         0.1s
```

c. vérifiez que l'image existe :

```
arij@server26:~/cv-onepage$ docker images | grep cv-onepage
cv-onepage:1.0      4d3efba7f495      94.7MB      26.8MB
```

d. Lancer le conteneur :

```
arij@server26:~/cv-onepage$ docker run -d --name cv-portfolio -p 8081:80 cv-onepage:1.0
3a8f6c145be18184629775a6be85aeb138ef5c6a24bc9a6300c895872baf07cd
```

E. Vérifiez que le conteneur tourne :

```
arij@server26:~/cv-onepage$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
3a8f6c145be1  cv-onepage:1.0 "/docker-entrypoint...." 26 seconds ago Up 25 seconds 0.0.0.0:8081->80/tcp, [::]:8081->80/tcp
v-portfolio
arij@server26:~/cv-onepage$
```

f. Tester l'accès depuis la VM :

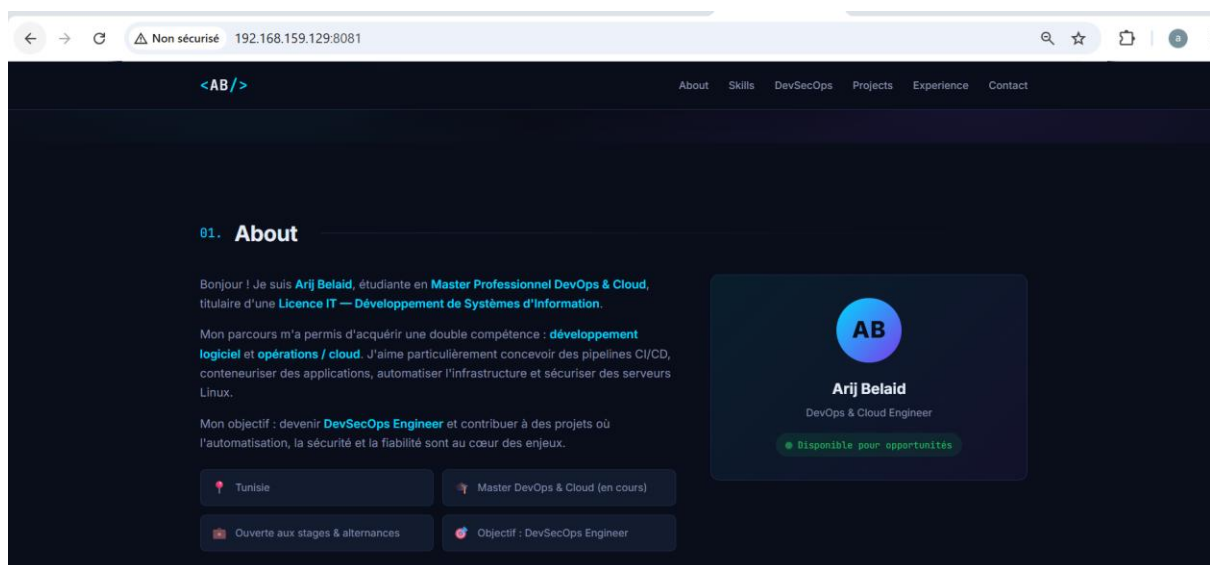
```
arij@server26:~/cv-onepage$ curl -I http://localhost:8081
HTTP/1.1 200 OK
Server: nginx
Date: Tue, 06 Oct 2026 09:59:41 GMT
Content-Type: text/html
Content-Length: 20201
Last-Modified: Tue, 06 Oct 2026 09:09:18 GMT
Connection: keep-alive
ETag: "6ac4babe-4ee9"
X-Frame-Options: SAMEORIGIN
X-Content-Type-Options: nosniff
X-XSS-Protection: 1; mode=block
Referrer-Policy: no-referrer-when-downgrade
Accept-Ranges: bytes
```

g. ouvrir le port 8081 dans UFW :

```
arij@server26:~/cv-onepage$ sudo ufw allow 8081/tcp
[sudo: authenticate] Password:
Rule added
Rule added (v6)
arij@server26:~/cv-onepage$ sudo ufw status
Status: active

To Action From
--
8080/tcp ALLOW Anywhere
22/tcp ALLOW Anywhere
8000/tcp ALLOW Anywhere
8081/tcp ALLOW Anywhere
8080/tcp (v6) ALLOW Anywhere (v6)
22/tcp (v6) ALLOW Anywhere (v6)
8000/tcp (v6) ALLOW Anywhere (v6)
8081/tcp (v6) ALLOW Anywhere (v6)
```

h. Tester depuis Windows



i. Commit + push

```
arij@server26:~/cv-onepage$ git add .
arij@server26:~/cv-onepage$ git commit -m "feat(v5): Dockerfile Nginx + capture du portfolio conteneurisé"
[main 2e21e37] feat(v5): Dockerfile Nginx + capture du portfolio conteneurisé
3 files changed, 72 insertions(+)
create mode 100644 .dockerignore
create mode 100644 Dockerfile
create mode 100644 nginx.conf
arij@server26:~/cv-onepage$ git push
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 2 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 1.48 KiB | 379.00 KiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:arijbelaidd/cv-onepage.git
10d1e68..2e21e37 main -> main
```

11- Construisez une image Docker nommée : cv-docker

README.md : ajoutez la commande utilisée et une capture d'écran du résultat.

a. Construire l'image cv-docker :

```
arij@server26:~/cv-onepage$ docker build -t cv-docker .
[+] Building 1.9s (12/12) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile              0.0s
=> => transferring dockerfile: 1.09kB                            0.0s
=> [internal] load metadata for docker.io/library/nginx:alpine  1.5s
=> [internal] load .dockerignore                                0.0s
=> => transferring context: 128B                                    0.0s
=> [1/7] FROM docker.io/library/nginx:alpine@sha256:df221db836e1754089190208cee7eeda94f233197056426eda74a43ab1abeac2 0.0s
=> => resolve docker.io/library/nginx:alpine@sha256:df221db836e1754089190208cee7eeda94f233197056426eda74a43ab1abeac2 0.0s
=> [internal] load build context                                0.0s
=> => transferring context: 157B                                    0.0s
=> CACHED [2/7] RUN rm -rf /usr/share/nginx/html/*              0.0s
=> CACHED [3/7] COPY index.html /usr/share/nginx/html/          0.0s
=> CACHED [4/7] COPY style.css /usr/share/nginx/html/           0.0s
=> CACHED [5/7] COPY script.js /usr/share/nginx/html/           0.0s
=> CACHED [6/7] COPY screenshot.png /usr/share/nginx/html/      0.0s
=> CACHED [7/7] COPY nginx.conf /etc/nginx/conf.d/default.conf  0.0s
=> exporting to image                                           0.1s
=> => exporting layers                                           0.0s
=> => exporting manifest sha256:2ac6b25f9fe460d03e4e0a6c849d30ffbe767be8cacd2cbe0521c93859bccd48 0.0s
=> => exporting config sha256:fb694bfff936746de84a4f705d8bf456eadaa5f42365eccc80018005b7aa0d637 0.0s
=> => exporting attestation manifest sha256:9c2bbdd30856564a754459a94962e5a892ea88530a69034f0a5582cc0faa0ac9 0.0s
=> => exporting manifest list sha256:fb69c5eb486f3ee7d78fcc72486cbf02498feffeee3c1b564feeb2f6fd58e7 0.0s
=> => naming to docker.io/library/cv-docker:latest              0.0s
=> => unpacking to docker.io/library/cv-docker:latest           0.0s
arij@server26:~/cv-onepage$
```

b. Vérifier que l'image existe

```
arij@server26:~/cv-onepage$ docker images | grep cv-docker
cv-docker:latest    fbb9c5eb486f    94.7MB    26.8MB    U
arij@server26:~/cv-onepage$
```

12- Exécutez un conteneur en exposant le Portfolio sur un port accessible et

vérifiez l'accès depuis la machine physique.

README.md : ajoutez la commande docker run, le résultat de docker ps et une capture d'écran.

a. Lancer un conteneur cv-docker sur le port 8082

```
arij@server26:~/cv-onepage$ docker run -d --name cv-docker-container -p 8082:80 cv-docker
b902662c64aa4a446bb2a83c5f104727dbc712cf5f82c4a1366567350b5d4883
arij@server26:~/cv-onepage$
```

b. Vérifier que le conteneur tourne

```
arij@server26:~/cv-onepage$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
AMES          b902662c64aa  cv-docker              "/docker-entrypoint..." 41 seconds ago Up 41 seconds 0.0.0.0:8082->80/tcp, [::]:8082->80/tcp c
v-docker-container
3a8f6c145be1  cv-onepage:1.0  "/docker-entrypoint..." 33 minutes ago Up 33 minutes 0.0.0.0:8081->80/tcp, [::]:8081->80/tcp c
v-portfolio
```

c. Ouvrir le port 8082 dans UFW

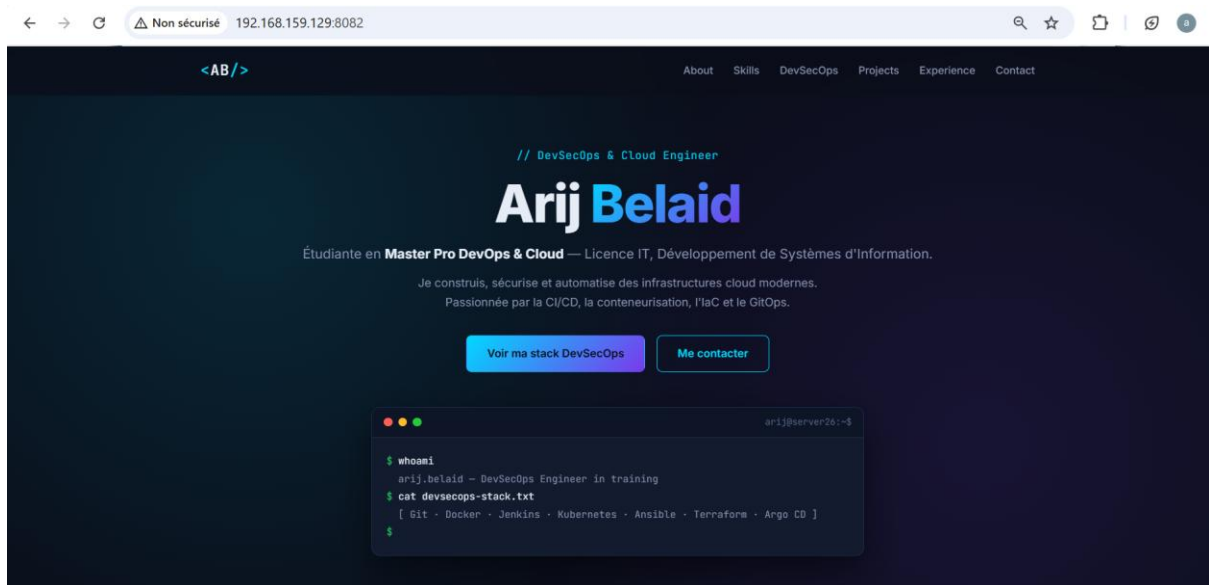
```
arij@server26:~/cv-onepage$ sudo ufw allow 8082/tcp
[sudo: authenticate] Password:
Rule added
Rule added (v6)
arij@server26:~/cv-onepage$ sudo ufw status
Status: active

To Action From
--
8080/tcp ALLOW Anywhere
22/tcp ALLOW Anywhere
8080/tcp ALLOW Anywhere
8081/tcp ALLOW Anywhere
8082/tcp ALLOW Anywhere
8080/tcp (v6) ALLOW Anywhere (v6)
22/tcp (v6) ALLOW Anywhere (v6)
8080/tcp (v6) ALLOW Anywhere (v6)
8081/tcp (v6) ALLOW Anywhere (v6)
8082/tcp (v6) ALLOW Anywhere (v6)
```

d. Tester en local (dans la VM)

```
arij@server26:~/cv-onepage$ curl -I http://localhost:8082
HTTP/1.1 200 OK
Server: nginx
Date: Tue, 06 Oct 2026 10:32:57 GMT
Content-Type: text/html
Content-Length: 20201
Last-Modified: Tue, 06 Oct 2026 09:09:18 GMT
Connection: keep-alive
ETag: "6ac4babe-4ee9"
X-Frame-Options: SAMEORIGIN
X-Content-Type-Options: nosniff
X-XSS-Protection: 1; mode=block
Referrer-Policy: no-referrer-when-downgrade
Accept-Ranges: bytes
```

e. Vérifier l'accès depuis la machine physique (Windows) :



f. Commit + Push :

```
arij@server26:~/cv-onepage$ git add .
arij@server26:~/cv-onepage$ git commit -m "feat(v7): exécution du conteneur cv-docker + accès depuis la machine physique"
[main ae7aa99] feat(v7): exécution du conteneur cv-docker + accès depuis la machine physique
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 screenshot2.png
arij@server26:~/cv-onepage$ git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 143.00 KiB | 1.18 MiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:arijbelaid/cv-onepage.git
2e21e37..ae7aa99 main -> main
arij@server26:~/cv-onepage$
```

13- Déployez le Portfolio avec Docker Compose : *docker compose up -d*

README.md : ajoutez la commande utilisée, le résultat de docker compose ps et une capture d'écran.

a. Créer le fichier docker-compose.yml :

```
arij@server26:~/cv-onepage$ nano docker-compose.yml
```

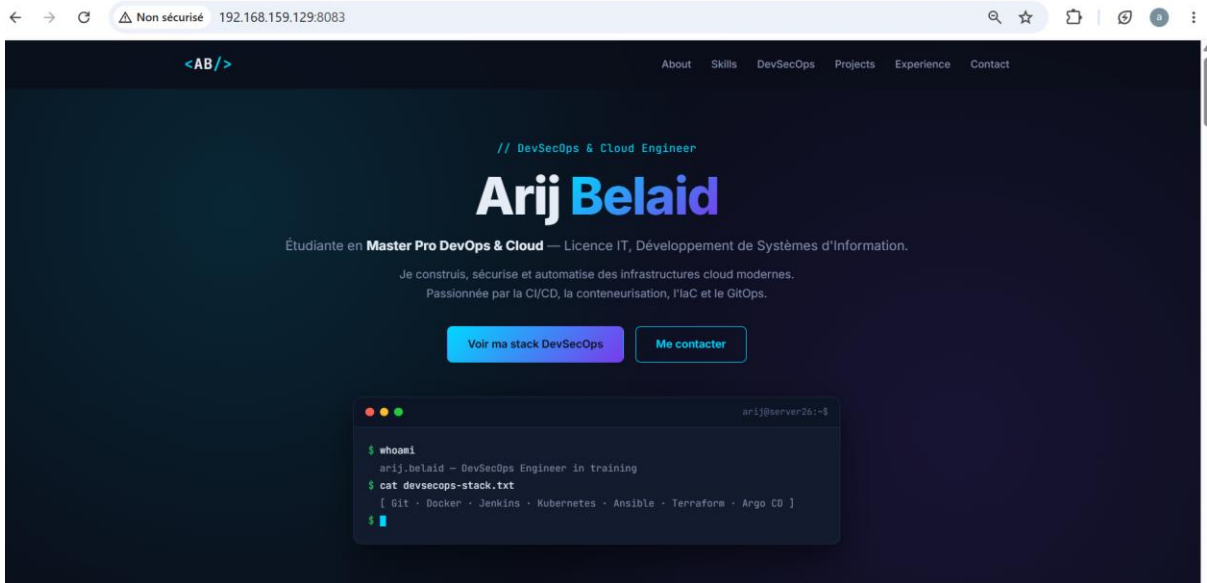
b. Lancer avec Docker Compose

```
arij@server26:~/cv-onepage$ docker compose up -d
[+] up 2/2
✓Network cv-onepage_default Created 0.2s
✓Container cv-portfolio-compose Started 1.0s
arij@server26:~/cv-onepage$
```

c. Vérifier avec docker compose ps

```
arij@server26:~/cv-onepage$ docker compose ps
NAME                IMAGE             COMMAND             SERVICE    CREATED         STATUS          PORTS
cv-portfolio-compose cv-docker:latest  "/docker-entrypoint..." portfolio    29 seconds ago  Up 28 seconds (healthy)  0.0.0.0:8080
```

Test :



14- Publiez les modifications du projet sur GitHub via SSH.

README.md : ajoutez les commandes Git utilisées et le lien du dépôt GitHub mis à jour.

a. Vérifier l'état Git

```
arij@server26:~/cv-onepage$ git status
git log --oneline -10
git remote -v
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
3834006 (HEAD -> main, origin/main) feat(v8): déploiement avec Docker Compose + healthcheck healthy
ae7aa99 feat(v7): exécution du conteneur cv-docker + accès depuis la machine physique
2e21e37 feat(v5): Dockerfile Nginx + capture du portfolio conteneurisé
10d1e68 feat(v4): section Projects générée dynamiquement en JavaScript depuis un tableau d'objets
1a281bf feat(v3): ajout section DevSecOps Skills (Git/Docker/Jenkins/K8s/Ansible/Terraform/Argo CD) + capture
f57ce9e feat(v2): évolution DevSecOps Portfolio + capture d'écran
a05a1e6 docs: ajout README avec commandes SSH et capture
a5ebb1b feat: CV one page Arij Belaid - HTML5/CSS3/JS
origin git@github.com:arijbelaid/cv-onepage.git (fetch)
origin git@github.com:arijbelaid/cv-onepage.git (push)
arij@server26:~/cv-onepage$
```

```
arij@server26:~/cv-onepage$ nano README.md
arij@server26:~/cv-onepage$ git add README.md
arij@server26:~/cv-onepage$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    modified:   README.md

arij@server26:~/cv-onepage$ git commit -m "docs: ajout section Git/SSH et lien dépôt (Question 14)"
[main 4df6e62] docs: ajout section Git/SSH et lien dépôt (Question 14)
 1 file changed, 76 insertions(+)
arij@server26:~/cv-onepage$ git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 1.38 KiB | 707.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To github.com:arijbelaid/cv-onepage.git
 3834006..4df6e62  main -> main
```

Première introduction à l'automatisation :

15- Installez Vagrant (dans la VM) et créez un premier Vagrantfile permettant de créer automatiquement une VM Ubuntu Server (vérifiez que votre VM dispose d'assez de RAM et d'espace disque).

a. Mettre à jour

```
arij@server26:~$ sudo apt update
[sudo: authenticate] Password:
Atteint : 1 http://tn.archive.ubuntu.com/ubuntu resolute InRelease
Réception de : 2 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Ign : 3 https://pkg.jenkins.io/debian-stable binary/ InRelease
Réception de : 4 http://tn.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Atteint : 5 https://pkg.jenkins.io/debian-stable binary/ Release
Réception de : 7 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [597 kB]
Réception de : 8 http://tn.archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Réception de : 9 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [137 kB]
Réception de : 10 http://tn.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [729 kB]
Réception de : 11 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46,7 kB]
Réception de : 12 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [474 kB]
Réception de : 13 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [96,3 kB]
Réception de : 14 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [199 kB]
Réception de : 15 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [64,8 kB]
Réception de : 16 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [55,5 kB]
Réception de : 17 https://download.docker.com/linux/ubuntu resolute InRelease [32,5 kB]
Réception de : 18 http://tn.archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [171 kB]
Réception de : 19 http://tn.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [97,7 kB]
Réception de : 20 http://tn.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [320 kB]
Réception de : 21 http://tn.archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [104 kB]
Réception de : 22 http://tn.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [201 kB]
Réception de : 23 http://tn.archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [1 056 B]
3 739 ko réceptionnés en 7s (561 ko/s)
21 paquets peuvent être mis à jour. Exécutez « apt list --upgradable » pour les voir.
arij@server26:~$
```

b. Clé GPG HashiCorp :

```
arij@server26:~$ ls -la /usr/share/keyrings/hashicorp-archive-keyring.gpg
-rw-r--r-- 1 root root 1213 Oct  6 21:12 /usr/share/keyrings/hashicorp-archive-keyring.gpg
```

c. ajout du dépôt HashiCorp +vérification :

```
arij@server26:~$ cat /etc/apt/sources.list.d/hashicorp.list
deb [signed-by=/usr/share/keyrings/hashicorp-archive-keyring.gpg] https://apt.releases.hashicorp.com resolute main
arij@server26:~$
```

d. Mettre à jour

```
arij@server26:~$ sudo apt update
Réception de : 1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Atteint : 2 http://tn.archive.ubuntu.com/ubuntu resolute InRelease
Atteint : 3 http://tn.archive.ubuntu.com/ubuntu resolute-updates InRelease
Atteint : 4 https://download.docker.com/linux/ubuntu resolute InRelease
Atteint : 5 http://tn.archive.ubuntu.com/ubuntu resolute-backports InRelease
Réception de : 6 https://apt.releases.hashicorp.com resolute InRelease [12,9 kB]
Ign : 7 https://pkg.jenkins.io/debian-stable binary/ InRelease
Réception de : 8 https://apt.releases.hashicorp.com resolute/main amd64 Packages [324 kB]
Réception de : 9 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [597 kB]
Atteint : 10 https://pkg.jenkins.io/debian-stable binary/ Release
Réception de : 12 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [137 kB]
1 209 ko réceptionnés en 2s (637 ko/s)
21 paquets peuvent être mis à jour. Exécutez « apt list --upgradable » pour les voir.
```

e. installer Vagrant

```
arij@server26:~$ sudo apt install -y vagrant
Installation de :
  vagrant

Sommaire :
  Mise à niveau de : 0. Installation de : 1, Supprimé : 0. Non mis à jour : 21
Taille du téléchargement : 90,6 Mo
Espace nécessaire : 274 Mo / 14,1 Go disponible

Réception de : 1 https://apt.releases.hashicorp.com resolute/main amd64 vagrant amd64 2.4.9-1 [90,6 MB]
90,6 Mo réceptionnés en 17s (5 431 ko/s)
Sélection du paquet vagrant précédemment désélectionné.
(Lecture de la base de données... 148296 fichiers et répertoires déjà installés.)
Préparation du dépaquetage de .../vagrant_2.4.9-1_amd64.deb ...
Dépaquetage de vagrant (2.4.9-1) ...
Paramétrage de vagrant (2.4.9-1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
```

f. Vérifier l'installation de Vagrant

```
arij@server26:~$ vagrant --version
Vagrant 2.4.9
arij@server26:~$
```

g. Installer VirtualBox :

```
arij@server26:~$ sudo apt install -y virtualbox
Installation de :
  virtualbox

Installation de dépendances :
  build-essential      libqt6dbus6
  bzip2                 libqt6gui6
  cpp                  libqt6help6
  cpp-15               libqt6network6
  cpp-15-x86-64-linux-gnu libqt6opengl6
  cpp-x86-64-linux-gnu libqt6printsupport6
  dkms                 libqt6qml6
  dpkg-dev             libqt6qmlmeta6
  fakeroot             libqt6qmlmodels6
  g++                  libqt6qmlworkerscript6
  g++-15              libqt6quick6
  g++-15-x86-64-linux-gnu libqt6sql6
  g++-x86-64-linux-gnu libqt6sql6-sqlite
  gcc                 libqt6statemachine6
  gcc-15              libqt6svg6
```

```
arij@server26:~$ VBoxManage --version
7.2.6_Ubuntu172322
arij@server26:~$
```

h. Créer le dossier + Vagrantfile :

```
arij@server26:~$ mkdir -p ~/vagrant-ubuntu
arij@server26:~$ cd ~/vagrant-ubuntu
arij@server26:~/vagrant-ubuntu$ nano Vagrantfile
```

```
Vagrant.configure("2") do |config|
  config.vm.box = "generic/ubuntu2204"
  config.vm.hostname = "vagrant-ubuntu"

  config.vm.provider "virtualbox" do |vb|
    vb.memory = 1024
    vb.cpus = 1
    vb.customize ["modifyvm", :id, "--hvvirtex", "off"]
    vb.customize ["modifyvm", :id, "--nestedpaging", "off"]
    vb.customize ["modifyvm", :id, "--paravirtprovider", "none"]
  end

  config.vm.provision "shell", inline: <<-SHELL
    apt-get update -y
    apt-get upgrade -y
  SHELL
end
```

Valider :

```
arij@server26:~/vagrant-ubuntu$ vagrant validate
Vagrantfile validated successfully.
```

16- Depuis la VM, connectez-vous à la VM créée par Vagrant avec : **vagrant ssh**
README.md : ajoutez une capture d'écran et comparez cette méthode avec la création manuelle de la VM.

a. Tenter vagrant ssh

```
Q16 - Connexion SSH à la VM Vagrant

=== 1. État de la VM Vagrant ===
Current machine states:

default                                not created (virtualbox)

The environment has not yet been created. Run 'vagrant up' to
create the environment. If a machine is not created, only the
default provider will be shown. So if a provider is not listed,
then the machine is not created for that environment.

=== 2. Tentative de connexion SSH ===
VM must be created before running this command. Run 'vagrant up' first.

=== 3. Rappel de la limitation ===
INFO: Your CPU does not support KVM extensions
INFO: For more detailed results, you should run this as root
HINT:   sudo /usr/sbin/kvm-ok
arij@server26:~/vagrant-ubuntu$
```

17- Modifiez le Vagrantfile pour configurer automatiquement :le nom de la VM

l'adresse IP privée, la mémoire et le nombre de CPU

README.md : ajoutez la configuration utilisée et le résultat de vagrant status.

a. Modifier le Vagrantfile :

```
GNU nano 8.7.1 Vagrantfile *
Vagrant.configure("2") do |config|

# -----
# Box Ubuntu Server
# -----
config.vm.box = "generic/ubuntu2204"

# -----
# 1. Nom de la VM (hostname)
# -----
config.vm.hostname = "vagrant-ubuntu"

# -----
# 2. Adresse IP privée
# -----
# Réseau privé accessible depuis la VM hôte uniquement
config.vm.network "private_network", ip: "192.168.56.10"

# -----
# 3. Provider VirtualBox
# -----
config.vm.provider "virtualbox" do |vb|

# Nom affiché dans VirtualBox
vb.name = "vagrant-ubuntu"

# 3. Mémoire RAM allouée (en Mo)
vb.memory = 1024
```

b. Valider le Vagrantfile :

```
arij@server26:~/vagrant-ubuntu$ vagrant validate
Vagrantfile validated successfully.
```

c. Tenter vagrant status :

```
arij@server26:~/vagrant-ubuntu$ vagrant status
Current machine states:

default                               not created (virtualbox)

The environment has not yet been created. Run 'vagrant up' to
create the environment. If a machine is not created, only the
default provider will be shown. So if a provider is not listed,
then the machine is not created for that environment.
arij@server26:~/vagrant-ubuntu$
```

Le contenu du Vagrantfile

```
Q17 - Vagrantfile configuré

# -*- mode: ruby -*-
# vi: set ft=ruby :

# =====
# Vagrantfile - VM Ubuntu Server (Question 17)
# Auteur : Arij Belaid
# =====
# Configuration automatique :
# - Nom de la VM       : vagrant-ubuntu
# - Adresse IP privée  : 192.168.56.10
# - Mémoire RAM        : 1024 Mo (1 Go)
# - Nombre de CPU      : 2 vCPU
# =====

Vagrant.configure("2") do |config|

# -----
# Box Ubuntu Server
# -----
config.vm.box = "generic/ubuntu2204"

# -----
# 1. Nom de la VM (hostname)
# -----
config.vm.hostname = "vagrant-ubuntu"

# -----
# 2. Adresse IP privée
# -----
```

```
# -----
# 2. Adresse IP privée
# -----
# Réseau privé accessible depuis la VM hôte uniquement
config.vm.network "private_network", ip: "192.168.56.10"

# -----
# 3. Provider VirtualBox
# -----
config.vm.provider "virtualbox" do |vb|

  # Nom affiché dans VirtualBox
  vb.name = "vagrant-ubuntu"

  # 3. Mémoire RAM allouée (en Mo)
  vb.memory = 1024

  # 4. Nombre de CPU (vCPU)
  vb.cpus = 2

  # Désactivation de la virtualisation imbriquée (limitation VT-x de l'hôte)
  vb.customize ["modifyvm", :id, "--hvmvrtex", "off"]
  vb.customize ["modifyvm", :id, "--nestedpaging", "off"]
  vb.customize ["modifyvm", :id, "--paravirtprovider", "none"]
end

# -----
# Provisionnement : mise à jour
# -----
config.vm.provision "shell", inline: <<-SHELL
apt-get update -y
apt-get upgrade -y
SHELL
```

Validation + Statut :

```
=====
Q17 - Validation et état
=====

=== vagrant validate ===
Vagrantfile validated successfully.

=== vagrant status ===
Current machine states:

default                                not created (virtualbox)

The environment has not yet been created. Run 'vagrant up' to
create the environment. If a machine is not created, only the
default provider will be shown. So if a provider is not listed,
then the machine is not created for that environment.
arij@server26:~/vagrant-ubuntu$
```